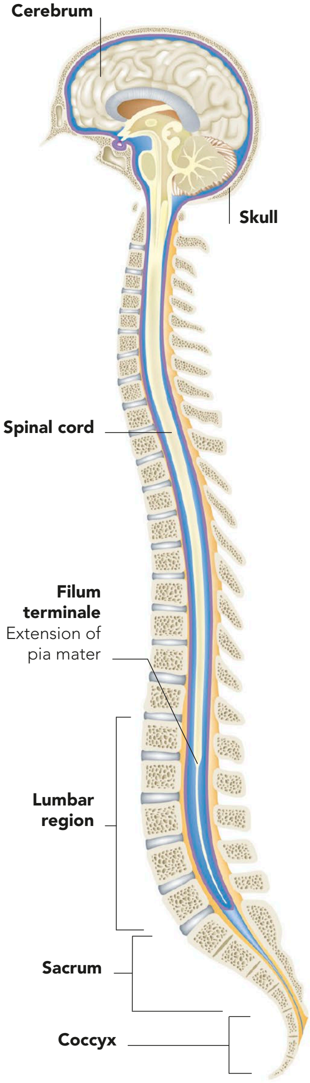
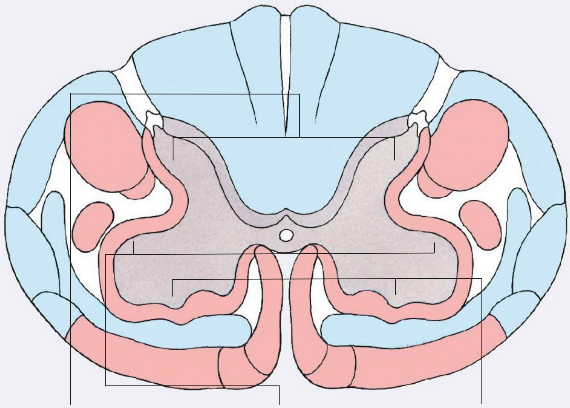
EXTENT OF THE SPINAL CORD

While the body is growing, the spinal cord does not continue to lengthen the way that the spinal bones do. By adulthood, it extends from the brain down to the first lumbar vertebra (L1) in the lower back. Here, the cord forms a cone-shaped ending that tapers to a slender, tail-like filament, known as the filum terminale. This extends down through the lumbar and sacral vertebrae to the coccyx.



NERVE TRACTS OF SPINAL CORD

In the white matter of the spinal cord, nerve fibres are grouped into main bundles, or tracts, according to the direction of the nerve signals that they carry and the type of signals they transmit and respond to, such as pain sensations or temperature. Some tracts connect and relay impulses between a few local pairs of spinal nerves, without sending fibres up to the brain. The central grey matter of the cord is organized into horns, or columns.



Dorsal (back) horns
Neurons here receive sensory information about touch, balance, muscle activity, and temperature

Lateral (side) horns
Neurons here monitor and regulate internal organs, such as the heart, lungs, stomach, and intestines

Ventral (front) horns
Neurons here send signals along motor fibres to skeletal muscles, causing them to contract and move

ASCENDING TRACTS
These ascending tracts are bundles of nerve fibres that relay impulses about bodily sensations, and inner sensors such as pain, up the spinal cord to the brain.

DESCENDING TRACTS
These descending tracts convey motor signals from the brain to the skeletal muscles of the torso and limbs in order to bring about voluntary movements.